📊 Power BI Project

Step-by-Step Student Guide

Business Information Systems (BIS)

This guide will walk you through every step — from choosing your topic to submitting your final report. Follow the steps in order and you will succeed!

📌 Step 1: Choose Your Project Topic

1. Your instructor will share an Excel file called `100_PowerBI_Project_Ideas.xlsx`.
2. Open the file and browse the **100 available project ideas** organized into categories like Retail, Finance, Healthcare, HR, Marketing, etc.
3. **Pick ONE idea** that interests your group.
4. Write your **Group Name** and **Student Names** in the empty columns next to the idea you chose.
5. ⚠️ **Important:** Each idea can only be taken by ONE group. First come, first served!

📌 Step 2: Find & Download Your Dataset

1. In the same Excel file, look at the column called **"Suggested Dataset Source"** — it will give you a link to start searching.
2. Go to the suggested website (usually **Kaggle.com** or **Google Dataset Search**).
3. Search for a dataset that matches your chosen project idea.
4. Download the file. It will usually be a `.CSV` or `.XLSX` file.

> 💡 ***Tip:*** *Choose a dataset with* ***at least 1,000 rows*** *and* ***5+ columns*** *(dates, categories, numbers). The more data, the better your dashboard will look.*

> 💡 ***Tip:*** *Kaggle requires a* ***free account*** *to download files. Sign up using your university email.*

📌 Step 3: Open Power BI & Import Your Data

1. Open **Power BI Desktop** on your computer (download it for free from Microsoft if you haven't already).
2. On the Home ribbon, click **"Get Data"**.
3. Choose:
   - **"Text/CSV"** if your file ends in `.csv`
   - **"Excel Workbook"** if your file ends in `.xlsx`
4. Select your downloaded file.
5. ⚠️ **DO NOT click "Load" yet!** Instead, click **"Transform Data"** — this opens the cleaning tool.

📌 Step 4: Clean Your Data (Power Query)

A new window called **Power Query Editor** will open. This is where you fix messy data before building charts.

What to do here:

1. **Remove useless columns:** Right-click on any column header that is empty or irrelevant → Click **"Remove"**.
2. **Remove blank rows:** Go to **Home** → Click **"Remove Rows"** → **"Remove Blank Rows"**.
3. **Fix data types:** Look at the small icon next to each column name:
   - 📅 Dates should show a **calendar icon** → If not, click the icon and change it to **"Date"**
   - 🔢 Numbers should show **"123"** → If not, change to **"Whole Number"** or **"Decimal Number"**
   - 🔤 Text should show **"ABC"**
4. When you are done cleaning, click **"Close & Apply"** (top-left corner).

> ✅ ***You are now back in Power BI with clean data!***

📌 Step 5: Build Your Data Model

A data model tells Power BI how your tables are connected to each other.

5a. Create a Date Table (Very Important!)

1. Go to the **Modeling** tab at the top.

2. Click **"New Table"**.
3. In the formula bar, type exactly:

```
Calendar = CALENDARAUTO()
```

4. Press **Enter**. You now have a Date Table!

5b. Connect Your Tables

1. Click the **Model View** icon (the 3rd icon on the left sidebar — it looks like connected boxes).
2. **Drag** the `Date` column from your new Calendar table onto the `Date` column in your main data table.
3. A line will appear connecting them — this means they are now related!

📌 Step 6: Write DAX Measures (Calculations)

Instead of dragging raw numbers into charts, you should create **Measures** — these are smart calculations.

How to create a Measure:

1. In the left panel, **right-click** on your main data table.
2. Click **"New Measure"**.
3. Type one of these formulas in the formula bar:

What you want	DAX Formula to type
Total Revenue	`Total Revenue = SUM(Sales[Revenue])`
Total Orders	`Total Orders = COUNTROWS(Sales)`
Average Price	`Avg Price = AVERAGE(Sales[Price])`
Profit Margin	`Profit Margin = DIVIDE(SUM(Sales[Profit]), SUM(Sales[Revenue]))`

> 💡 ***Tip:*** *Replace `Sales` with YOUR table name, and `Revenue` / `Price` / `Profit` with YOUR column names.*

📌 Step 7: Build Your Dashboard (Charts & Visuals)

Now the fun part! Go back to **Report View** (the 1st icon on the left sidebar).

Follow this order:

7a. Add KPI Cards (Top of the page)

1. Click the **Card** visual from the Visualizations panel.
2. Drag your DAX Measure (e.g., `Total Revenue`) into it.
3. Repeat for 3–4 key numbers. Place them in a row at the top.

7b. Add Charts (Middle of the page)

What you want to show	Use this chart
How something changes over time	📈 **Line Chart**
Compare categories (Top 10 Products, Sales by City)	📊 **Bar Chart** or **Column Chart**
Show proportions (Male vs Female, Yes vs No)	🍩 **Donut Chart** (use only for 2–5 categories!)
Show locations on a map	🗺️ **Map Visual**

7c. Add Slicers (Interactive Filters)

1. Click the **Slicer** visual.

2. Drag a column like `Year`, `Region`, or `Department` into it.
3. Now anyone viewing your dashboard can click the slicer to filter the entire page!

📌 Step 8: Write Your Report on Overleaf (LaTeX)

You must submit a professional written report using **LaTeX** on **Overleaf**.

8a. Setup

1. Go to www.overleaf.com and create a **free account**.
2. Click **"New Project"** → **"Upload Project"**.
3. Upload the template file your TA gave you: **`powerbi_project_template.tex`**

8b. Fill in the Template

The template already has all the sections. Just replace the placeholder text `[...]` with your own content:

Section	What to write
Abstract	2–3 sentences: What business problem did you solve? What did you find?
Dataset Description	Where did you get the data? How many rows/columns? What are the key variables?
Data Cleaning (ETL)	What did you fix in Power Query? (Removed nulls? Changed data types?)
DAX Code	Copy-paste your DAX formulas into the code blocks provided in the template.
Dashboard Screenshots	Take screenshots of your dashboard → Upload them to Overleaf → Insert them using `\includegraphics`. For each screenshot, write 2–3 sentences explaining **what business insight it shows**.
Conclusion	What should the company DO based on your data? Give 2–3 recommendations.

8c. Export Your Report

1. Click the **"Recompile"** button (or it compiles automatically).
2. Click **"Download PDF"** to save your final report.

📌 Step 9: Prepare Your Presentation

When you present your project, you are NOT presenting a school assignment. **You are an analyst presenting to the CEO of a company.**

❌ **Don't say:** "*I used a bar chart and a slicer.*"

✅ **Do say:** "*Our analysis shows that sales in Region X dropped 20% in Q3. We recommend increasing the marketing budget in that area.*"

Presentation Tips:

- Keep it to **5 minutes** per group.
- Show your **live dashboard** on screen (open Power BI, not just screenshots).
- Click your slicers during the presentation to show interactivity.
- End with **2–3 clear business recommendations**.

📌 Step 10: Submit Your Work

You must submit TWO files:

File	Format	How to get it
Your Dashboard	`.PBIX`	Save from Power BI Desktop → File → Save As
Your Written Report	`.PDF`	Download from Overleaf

> ⚠️ ***Warning:*** *Submitting a Word document instead of a LaTeX PDF will result in a grade deduction. Use the Overleaf template!*

🎯 Quick Summary

Step 1: Pick your topic → **Step 2:** Download data → **Step 3:** Import into Power BI

Step 4: Clean in Power Query → **Step 5:** Build data model → **Step 6:** Write DAX

Step 7: Build dashboard → **Step 8:** Write report on Overleaf → **Step 9:** Present → **Step 10:** Submit!